Full matched study with optimized context text

Repeated sentence description | 30 October 2026

Summary

The original matched cohort was fixed at 100 candidate and 100 control variant–gene pairs. Both descriptions were scored across all four blood-cell types wherever an observed beta was available: 642 observations per description, including 321 candidate-derived and 321 control-derived observations. There were 158 unavailable variant–gene–cell estimates per description out of 800 possible; these remain exclusions, not zero effects.
Across all available candidate-derived observations, direction agreement changed from 65.0% (95% locus CI 56.0–73.4%) to 69.5% (CI 60.4–77.8%). Pooled Spearman rho changed from 0.328 to 0.371. Correct directions per all available observations changed from 156/321 to 166/321; paired gain 3.1 pp (95% CI -1.2 to 7.6 pp). Repeated cells were clustered by locus. These are prompt-tuned, observational results, not an independent zero-shot replication.

Original discovery-cell pilot repeated

Condition	All	Nonzero	Sign agreement / 95% CI	Pooled Spearman rho
baseline candidate	100	77	67.5% / 56.4–77.9%	0.370
baseline control	100	60	51.7% / 39.3–64.4%	-0.006
best candidate	100	80	75.0% / 65.4–84.6%	0.442
best control	100	57	50.9% / 38.9–63.0%	0.033

Full scoring across available cells

Condition	All	Nonzero	Sign agreement / 95% CI	Pooled Spearman rho
baseline candidate	321	240	65.0% / 56.0–73.4%	0.328
baseline control	321	175	48.6% / 39.7–57.8%	-0.026
best candidate	321	239	69.5% / 60.4–77.8%	0.371
best control	321	187	42.8% / 34.0–51.4%	-0.070

Methods, scope and interpretation

The expansion gate required both discovery-candidate correct/all-pair count and Spearman correlation to improve. The frozen repeated description passed this gate. No further prompt or variant selection used the full-scoring results. The source cohorts, control matches, allele orientation, TSSs, model and scored windows were identical across descriptions.

Chosen description

Homo sapiens peripheral-blood {cell}, unstimulated. Single-cell RNA sequencing with 10x Genomics Chromium 3-prime poly(A) capture; source material: peripheral blood mononuclear cells (PBMCs). Homo sapiens peripheral-blood {cell}, unstimulated. Single-cell RNA sequencing with 10x Genomics Chromium 3-prime poly(A) capture; source material: peripheral blood mononuclear cells (PBMCs).

Cell specificity

baseline: 412 pairwise contrasts, 330 non-abstaining; sign agreement 54.5% (95% CI 47.9–60.7%); Spearman rho 0.172. Contrasts share variants, genes and donors; covariance of observed cell effects remains unknown, so these are descriptive model comparisons, not tests of biological heterogeneity.
best: 412 pairwise contrasts, 335 non-abstaining; sign agreement 57.0% (95% CI 51.1–62.8%); Spearman rho 0.252. Contrasts share variants, genes and donors; covariance of observed cell effects remains unknown, so these are descriptive model comparisons, not tests of biological heterogeneity.
The paired cell-contrast gain in correct predictions per all contrasts was 2.7 pp (95% locus CI -4.1 to 9.7 pp). Individual above-chance concordance does not by itself demonstrate improvement over the baseline.

Limitations and comparison

The text was selected using development outcomes in earlier prompt searches. That original-cohort discovery gain was partly in-sample. The prior new-locus check gave 10/20 correct for both original and selected text; thus independent improvement was not established before this expansion. Full scoring adds cell contexts to the same selected loci, rather than adding an independent cohort. Candidate-derived comparison-cell associations need not have PIP≥0.9 or be significant outside the discovery cell. Controls remain uncertain comparators; absent PIP is not zero.
The model g0-expression-8192 used 40,960-bp oriented inputs and exact matched REF/ALT effective windows, as documented in the earlier geometry amendment. Minus-strand edits preserve ALT-minus-REF effect orientation. The initial 81,920-bp geometry was not restored. Scores are log(TPM+1), while SAIGE-QTL betas are count-model effects. GI training overlap and an immutable deployed weight revision remain unresolved. Gene-window clusters approximate LD. No causal validation is implied.
Cuomo et al., Nature (2026), doi:10.1038/s41586-026-10721-2, reports PromoterAI median correlation 0.66 on lead eQTLs. Our pooled Spearman coefficients use different variants, selection and aggregation; they cannot rank the models without running them on matched inputs and observations.

Execution and reproducibility

Completed all available cells: True. Stored prediction records: 1284. New requests this repeat/expansion: 1340; cumulative prediction attempts including the initial preledger control: 2784. The API developer/user confirmed quota lifted for this key. Requests remained serial with a 4,000-attempt software cap, no automatic retries and logged rate headers. No monetary price was available. Small measured tables, source hash, all missing-data exclusions, both descriptions and metrics accompany this report. Original benchmark outputs remain unchanged.
Reproduce with GI_API_KEY exported: python scripts/full_best_description.py; python scripts/report_full_best.py. Model calls resume from the content-addressed cache. Data: Zenodo 18221260; GRCh38/GENCODE v44. Offline testing result is in test_results.txt.